

Circle Packing

A Visual Introduction

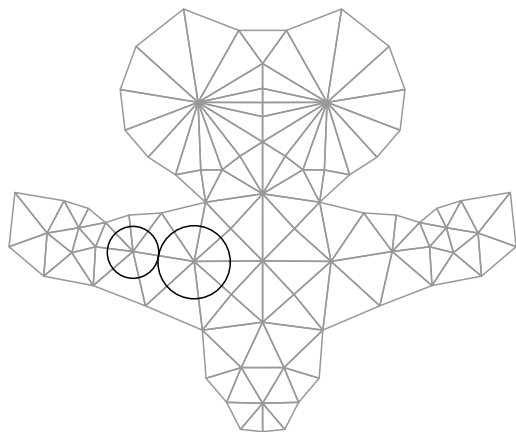
James Ashe

Western Governors University

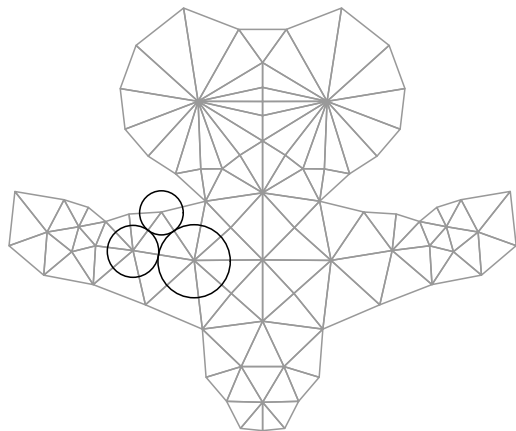


What is a Circle Packing?

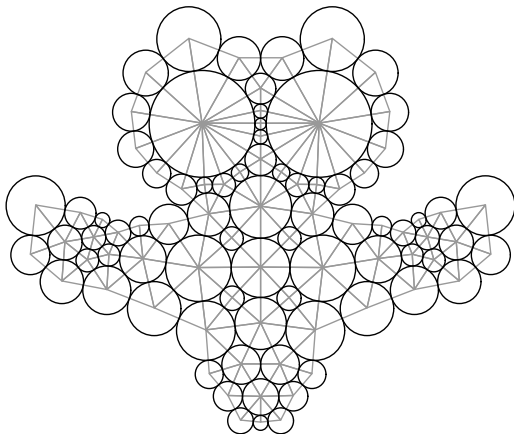
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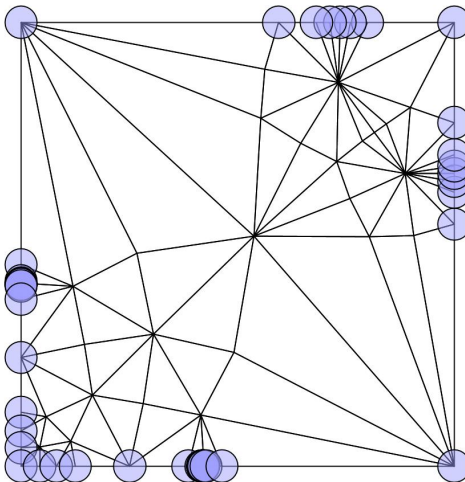


A collection of circles with a pattern such that

1. The circles have one and only one circle, C_i , associated with each vertex v_i in the pattern.
2. Any two neighbors in the circle collection are *tangent*.
3. $\{C_i, C_j, C_k\}$ form a positively oriented triple whenever $\{v_i, v_j, v_k\}$ form a positively oriented face in the pattern.

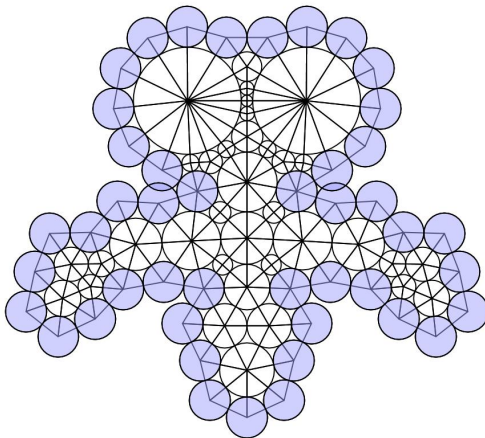
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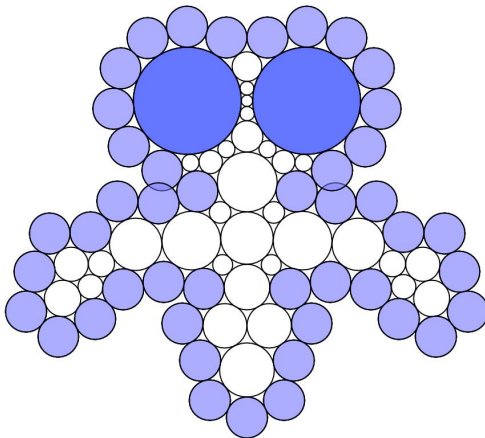
What is a Circle Packing?

Complex + Circles = Geometry



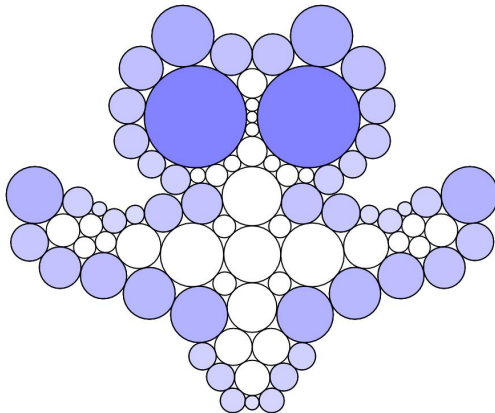
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- Border conditions introduce flexibility.

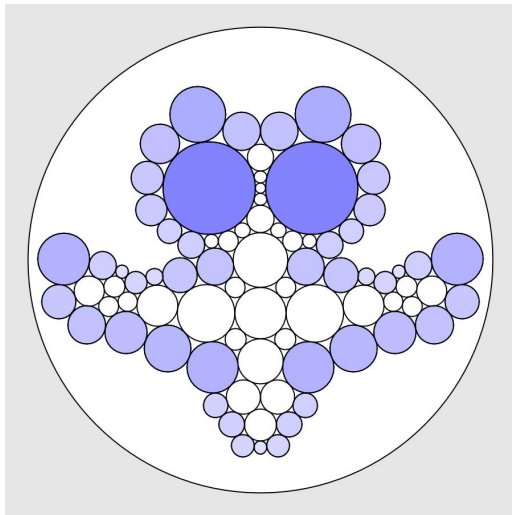
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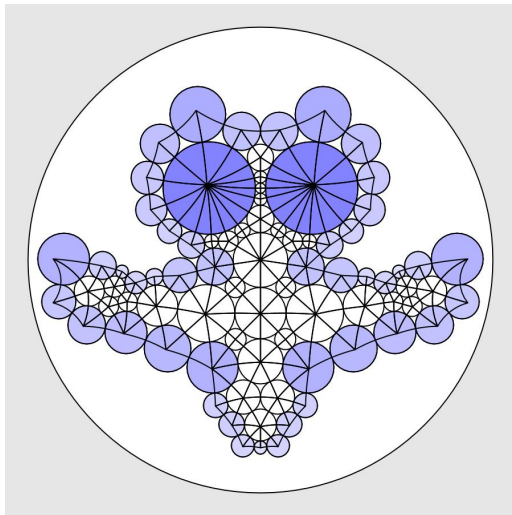
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- The plane, $\mathbb{C}$.
- The disc, $\mathbb{D}$.



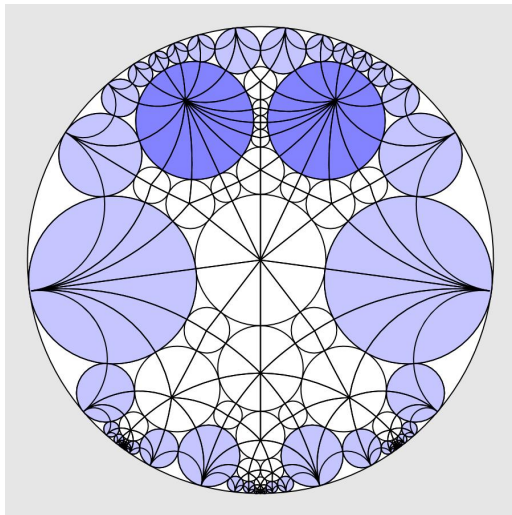
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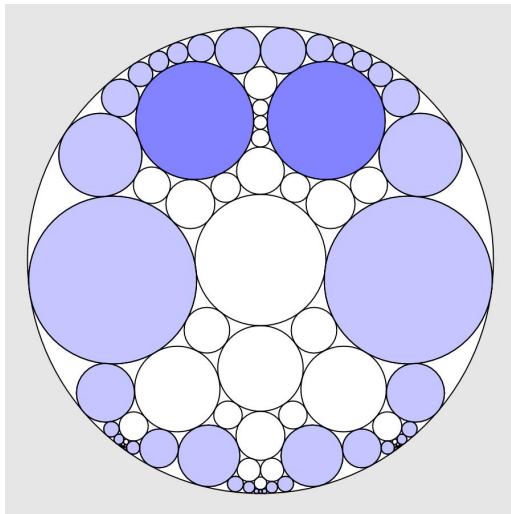
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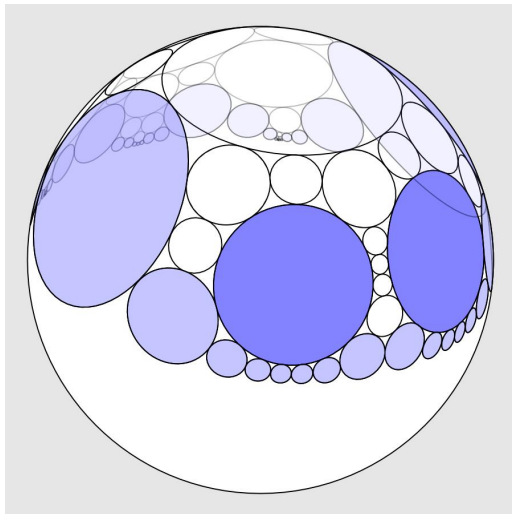
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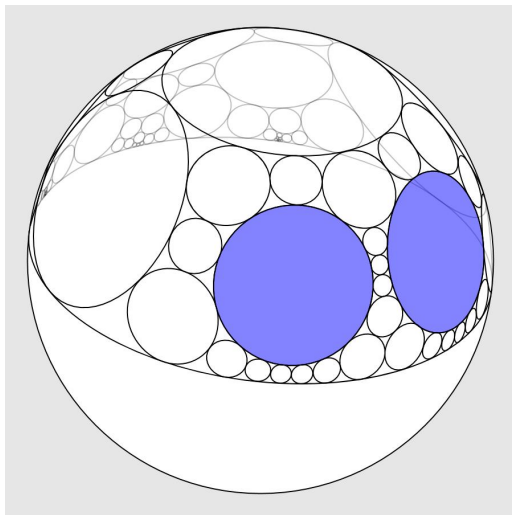
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- The plane, $\mathbb{C}$.
- The disc, $\mathbb{D}$.
- The sphere, $\mathbb{P}$.



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Moving between Geometries

Via Stereographic Projections of the Riemann sphere.

- A Möbius transformation

Moving between Geometries

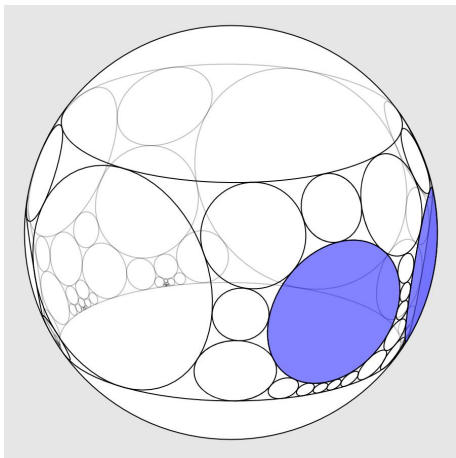
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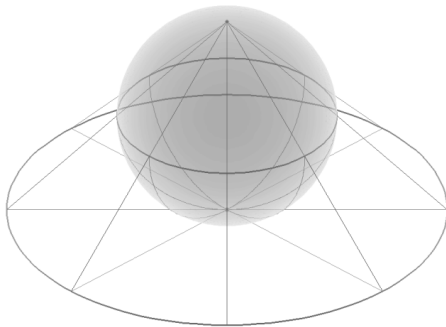
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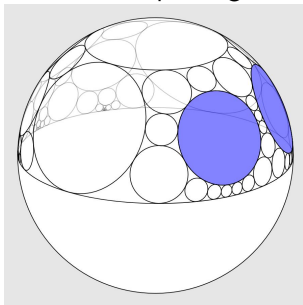
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There exists a CP on the sphere for any finite triangulation of the sphere. Furthermore, this CP is unique up to Möbius transformations.

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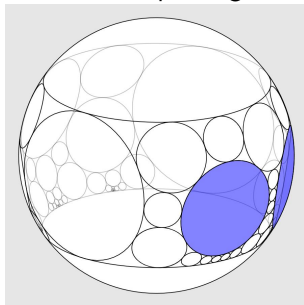
When does a circle packing on $\mathbb{P}$ exist?



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When does a circle packing on $\mathbb{P}$ exist?

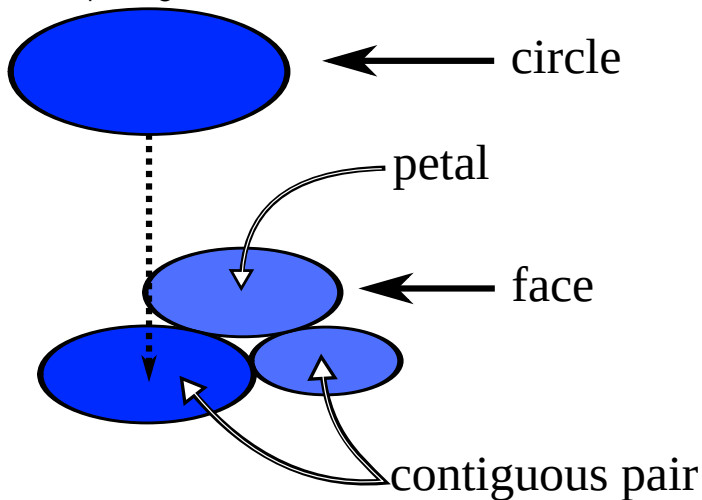


Preliminaries for KAT Proof

Hierarchy of a circle packing

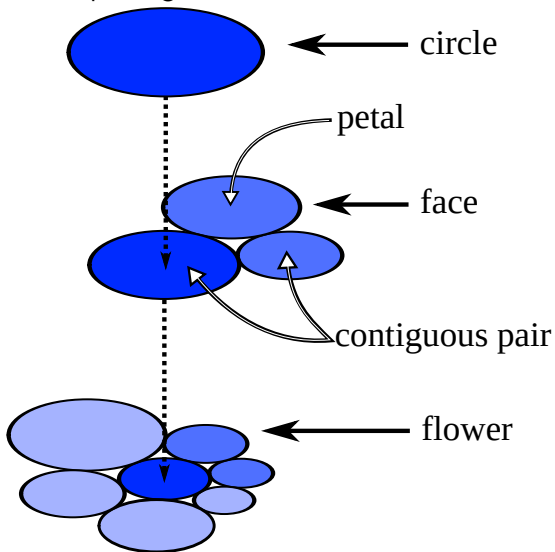
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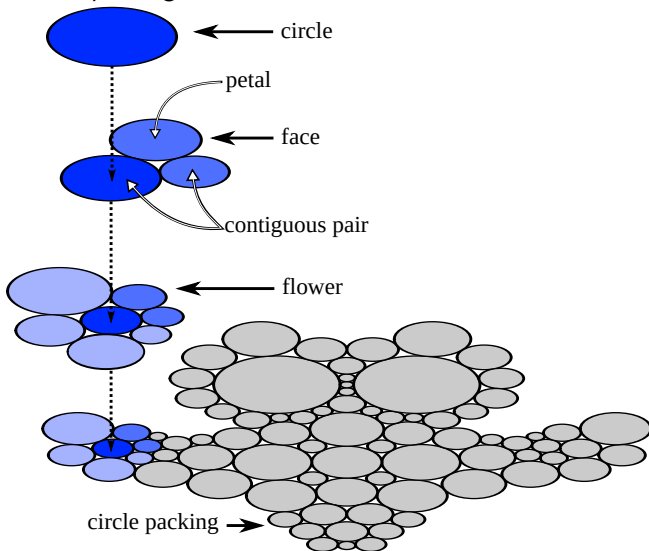
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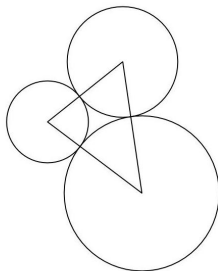
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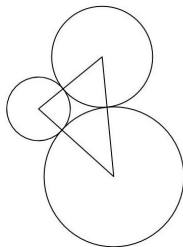
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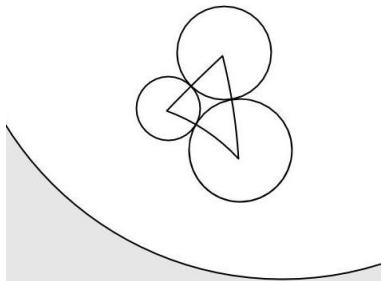
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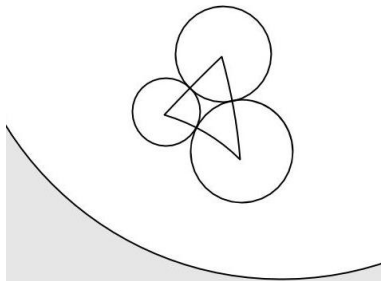
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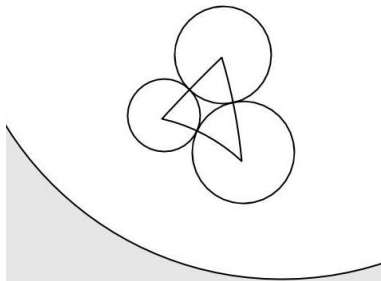
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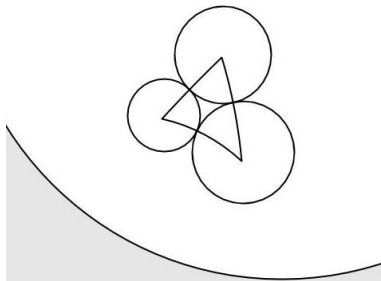
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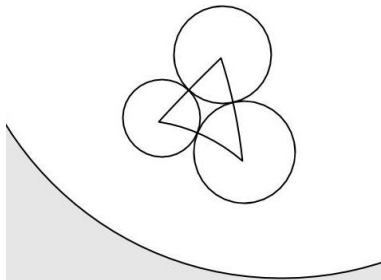
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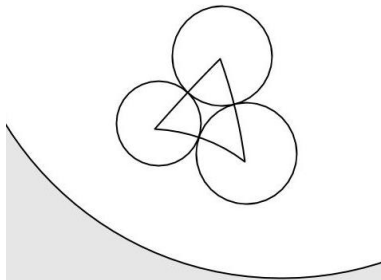
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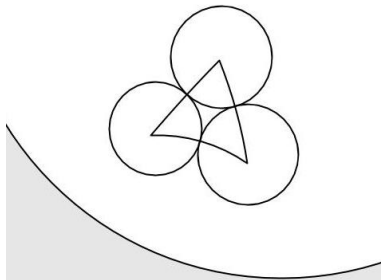
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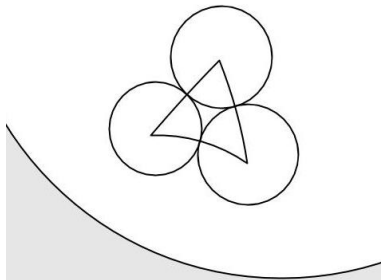
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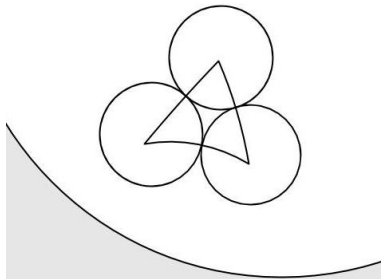
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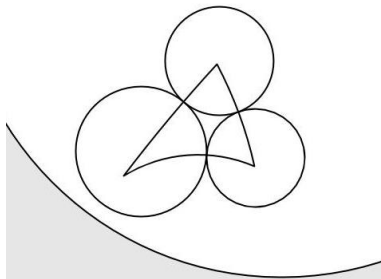
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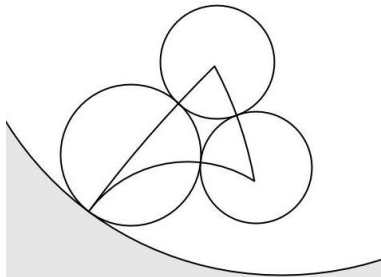
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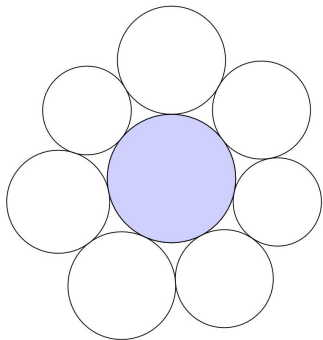
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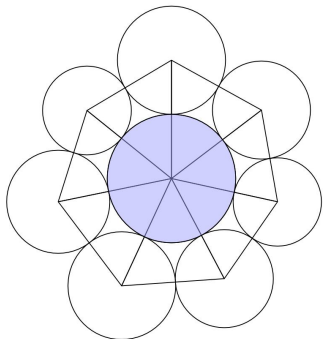
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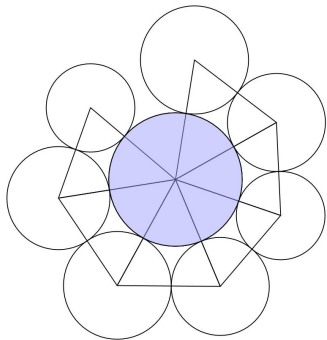
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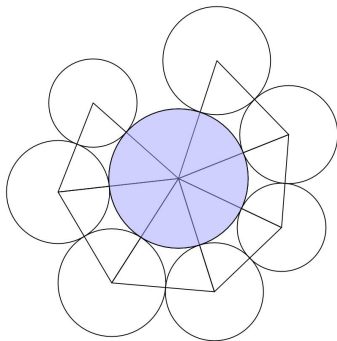
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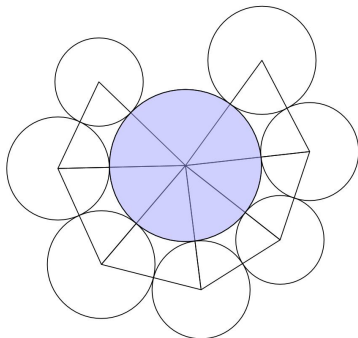
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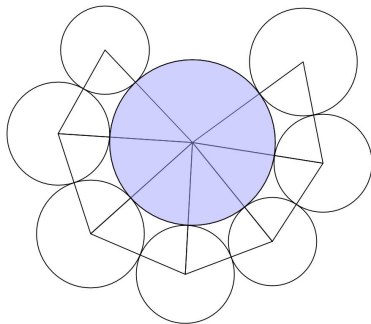
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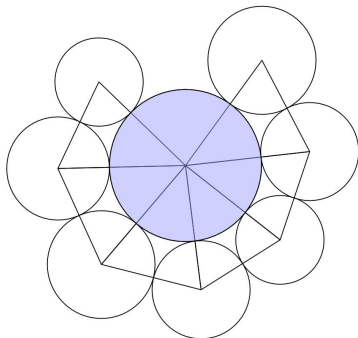
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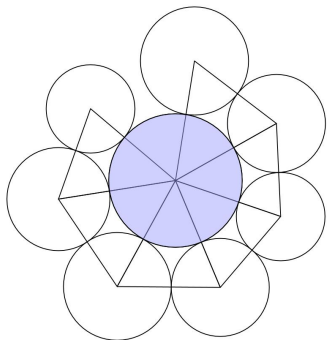
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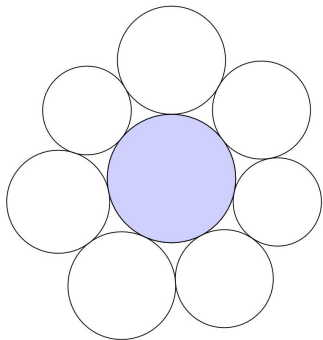
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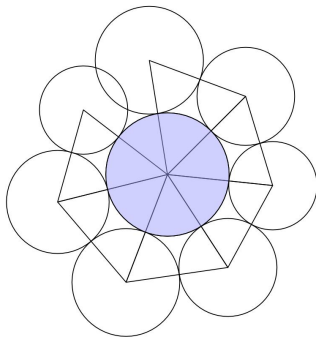
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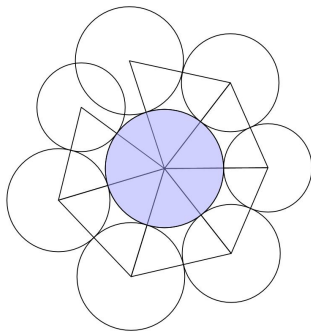
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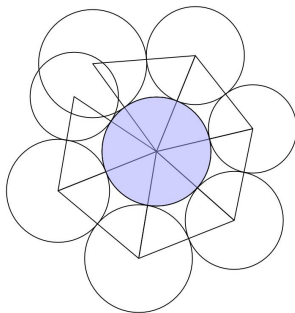
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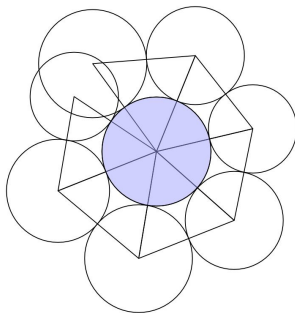
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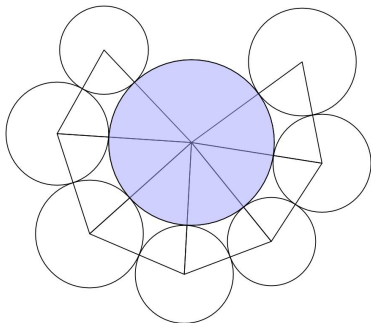
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- *Packing condition:* Angle sum must be $2\pi n$.

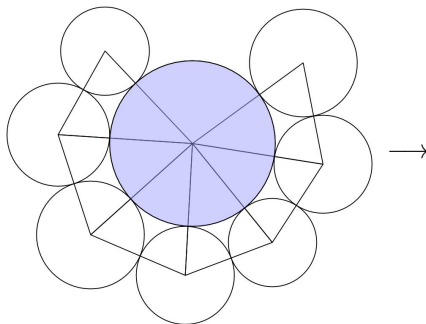
Let r_i be the radius at v_i . Define a **radii assignment function**, $\mathfrak{R}$

$$\mathfrak{R}(r_i) = \begin{cases} \text{set } r_i \text{ so the angle sum at } v_i \text{ is } 2\pi & \text{if } v_i \in \text{interior} \\ \text{unchanged} & \text{if } v_i \in \text{boundary} \end{cases}$$



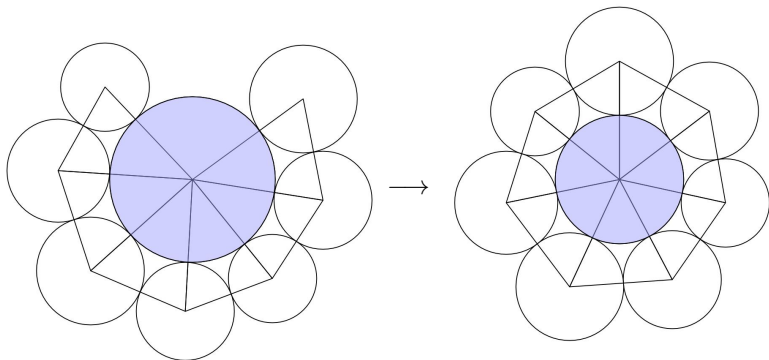
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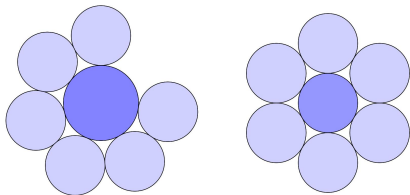
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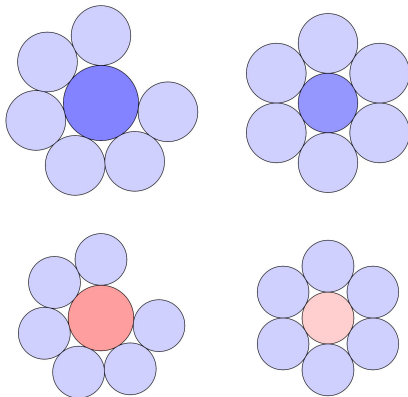
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argument $\Rightarrow$ uniqueness.

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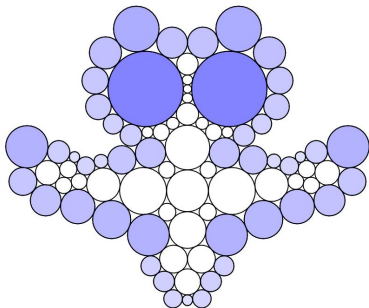
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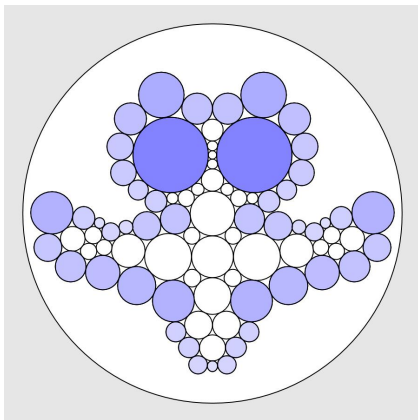
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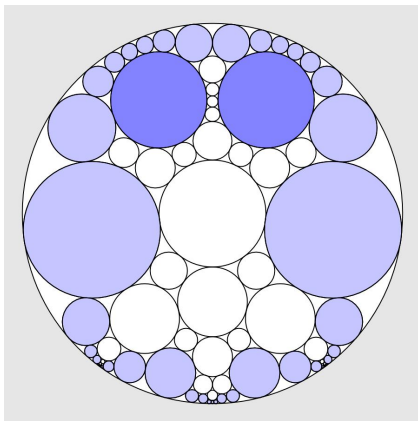
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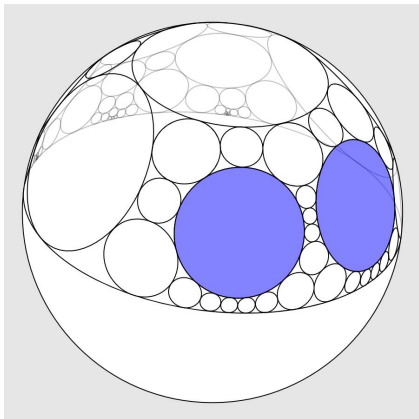
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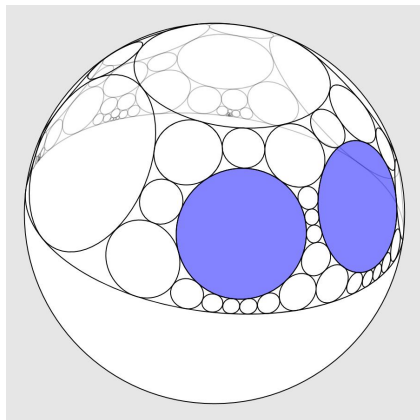
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Koebe-Andreev-Thurston Theorem (KAT)

There exists a CP on the sphere for any finite triangulation of the sphere. Furthermore, this CP is unique up to Möbius transformations.



Circle Packings as Discrete Analytical Function

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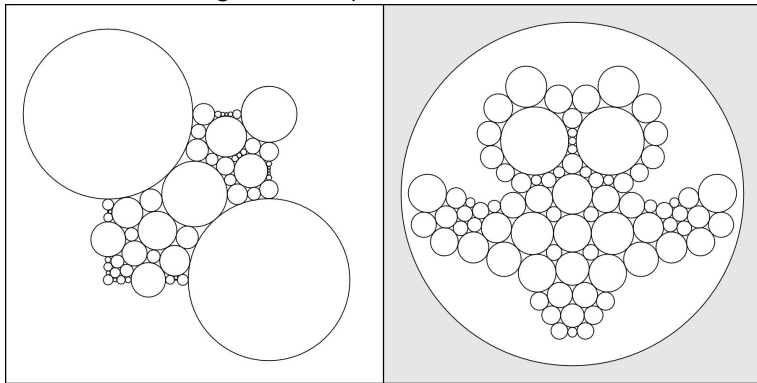
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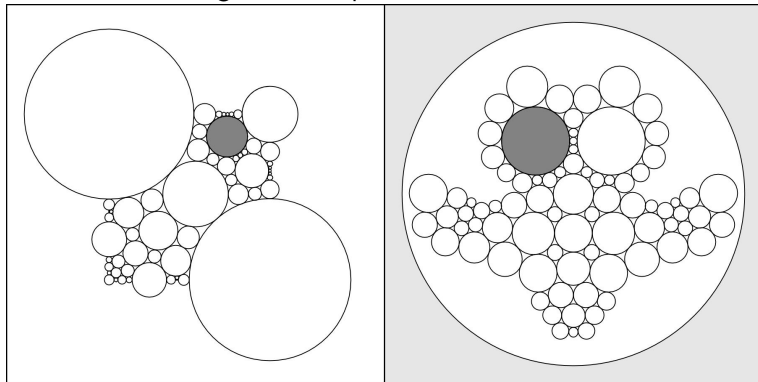
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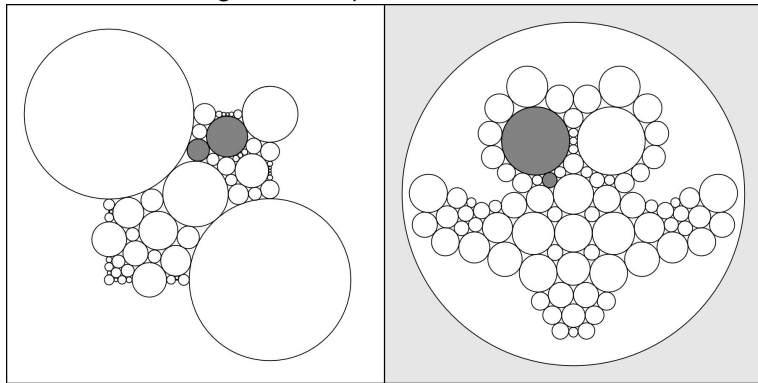
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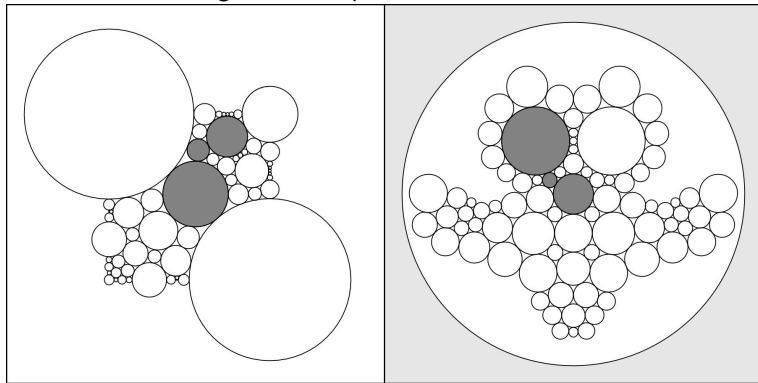
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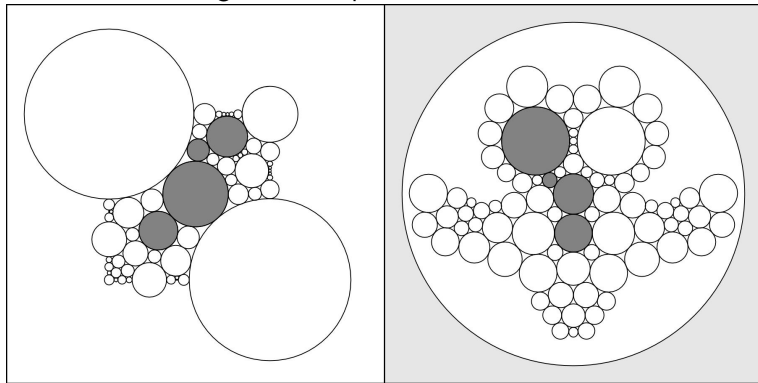
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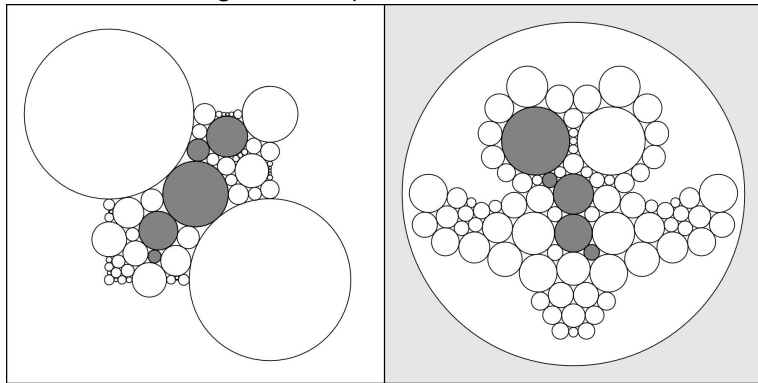
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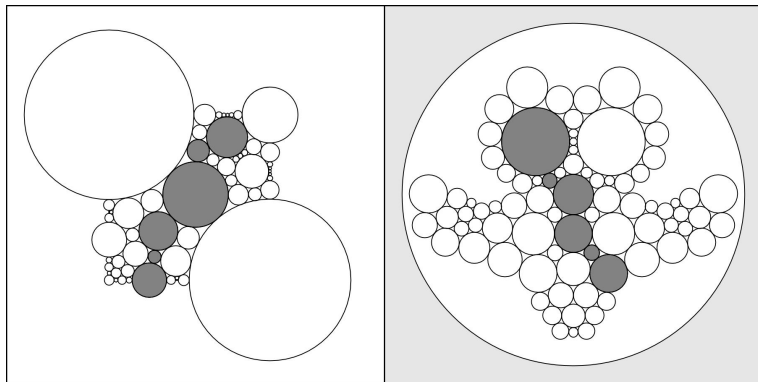
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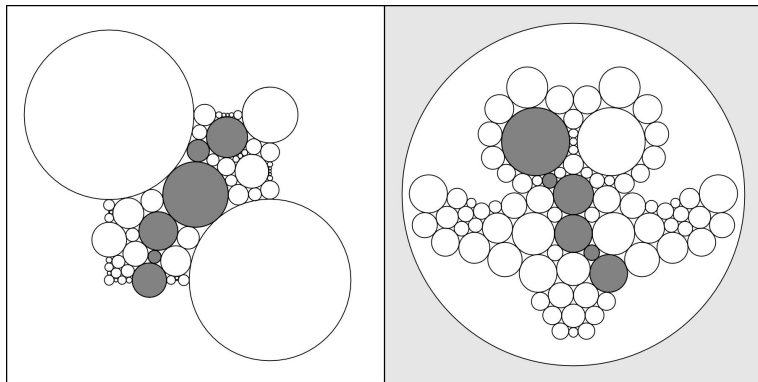
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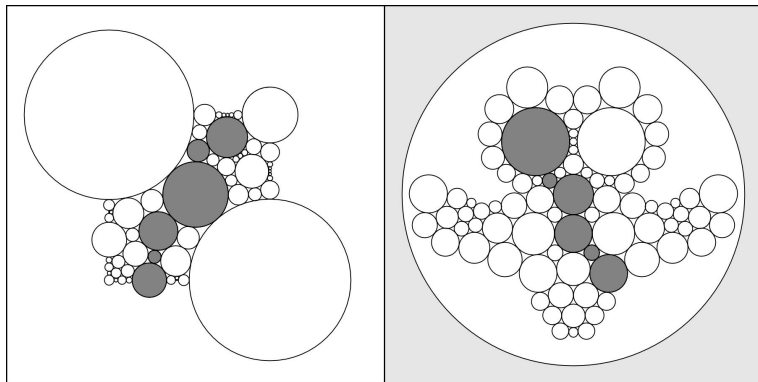
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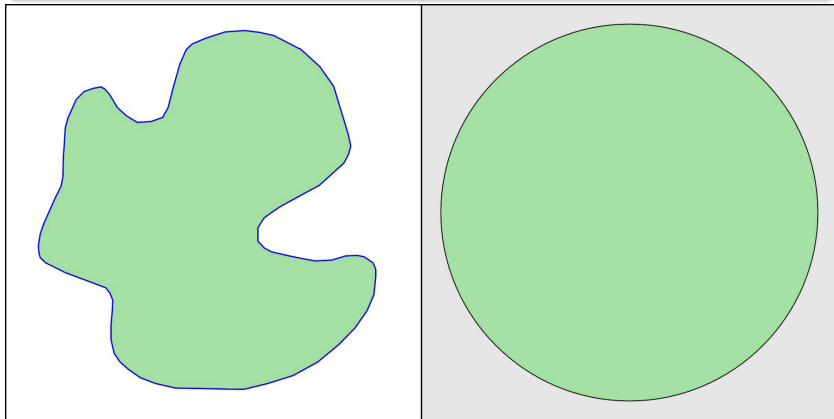


Analytic functions, *map infinitesimal circles to infinitesimal circles.*
Circle packing, *maps real circles to real circles.*

Riemann Mapping Theorem

Classical RMT

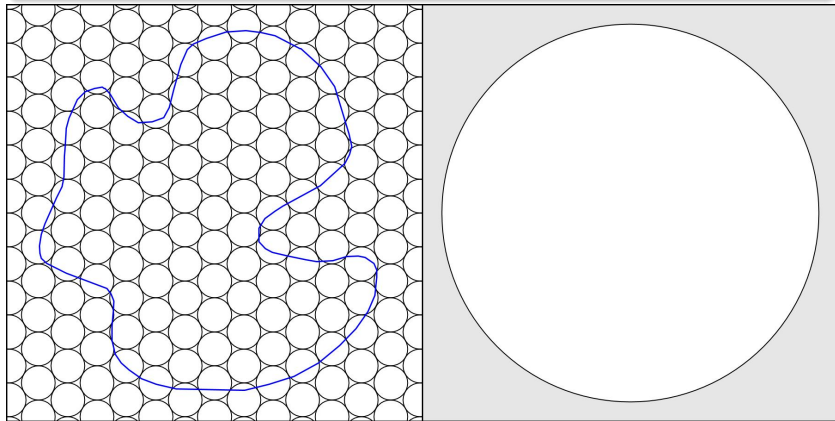
If R is a non-empty simply connected proper subset of $\mathbb{C}$, then there exists an essentially unique mapping from R onto the unit disc.



Riemann Mapping Theorem

Discrete RMT

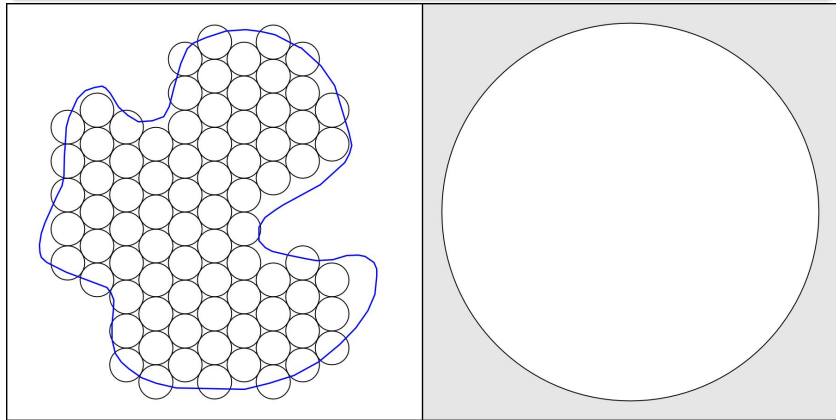
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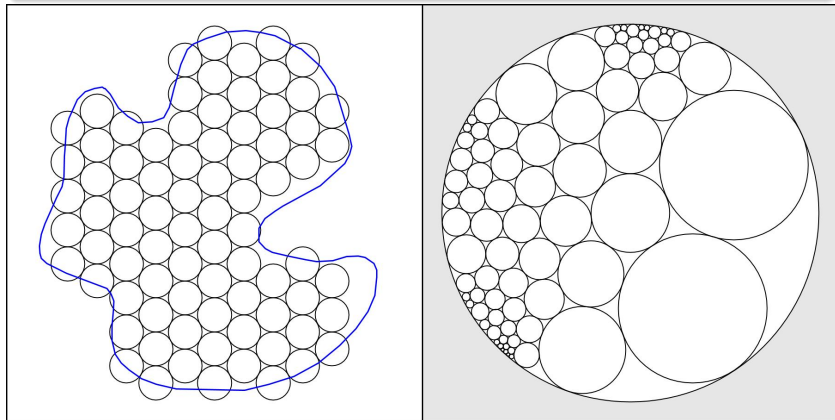
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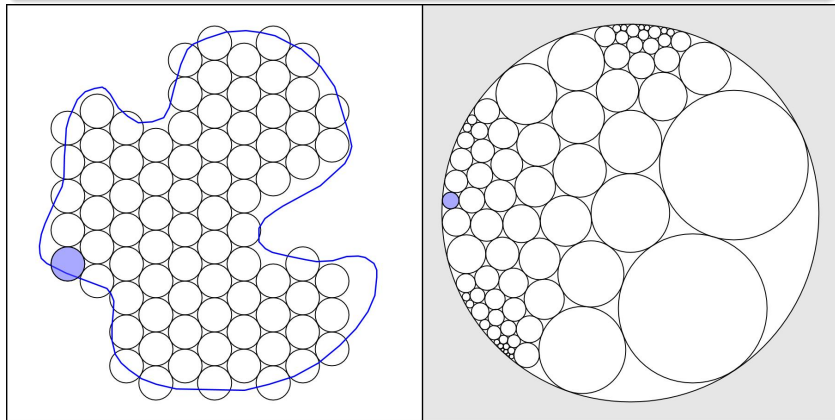
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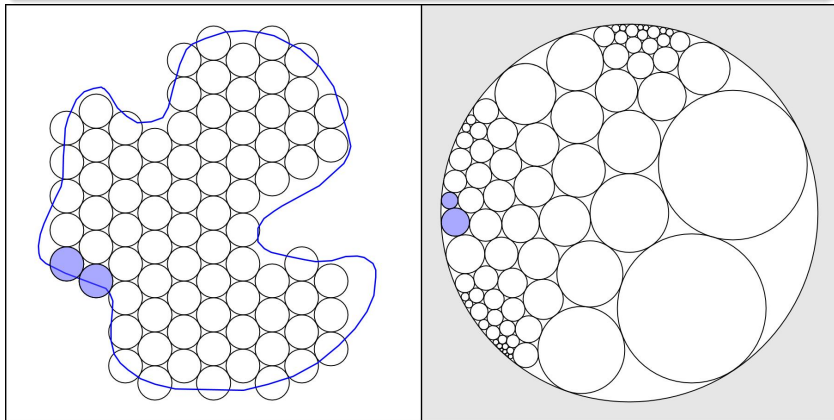
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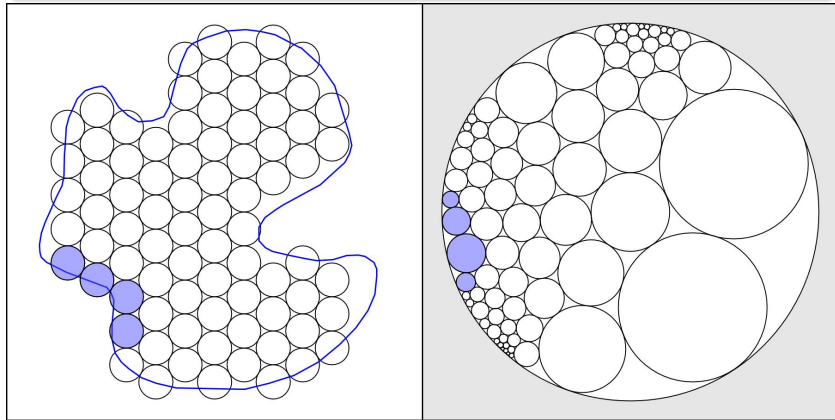
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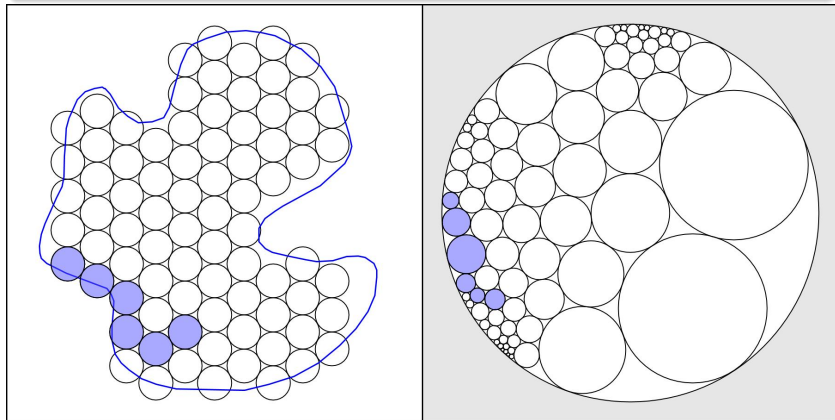
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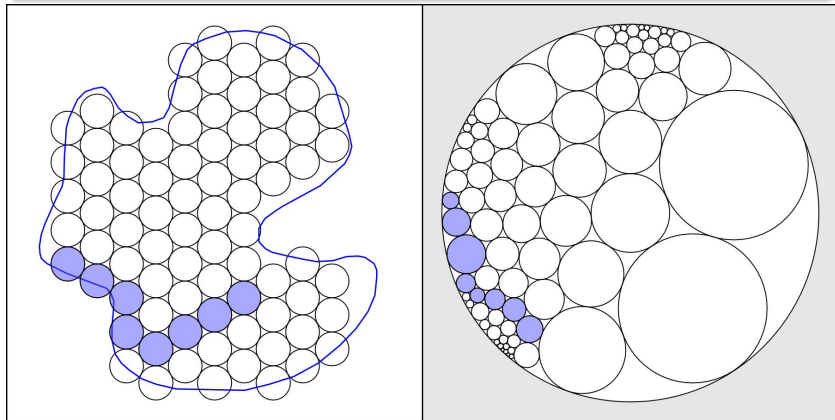
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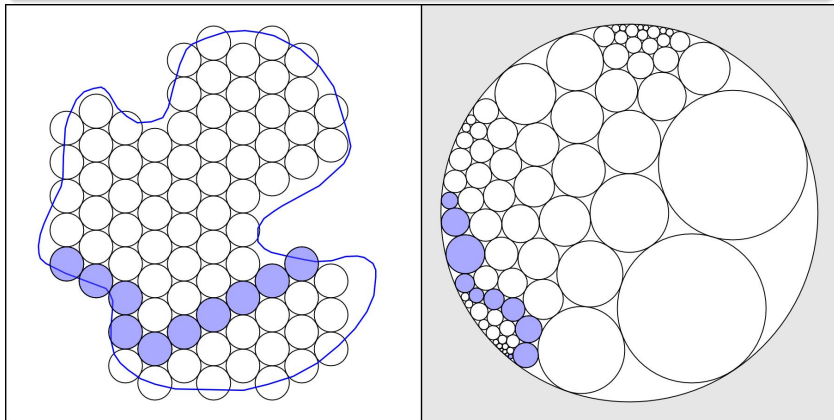
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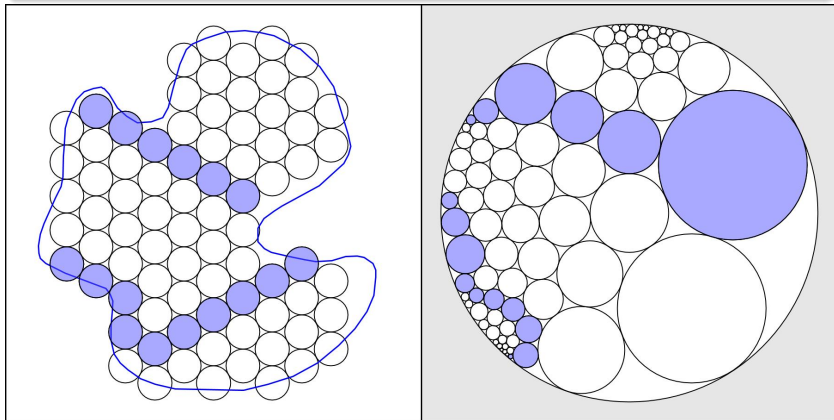
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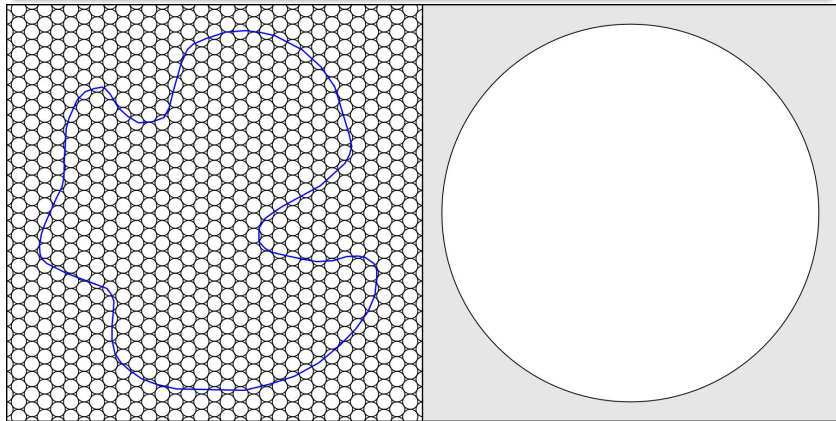
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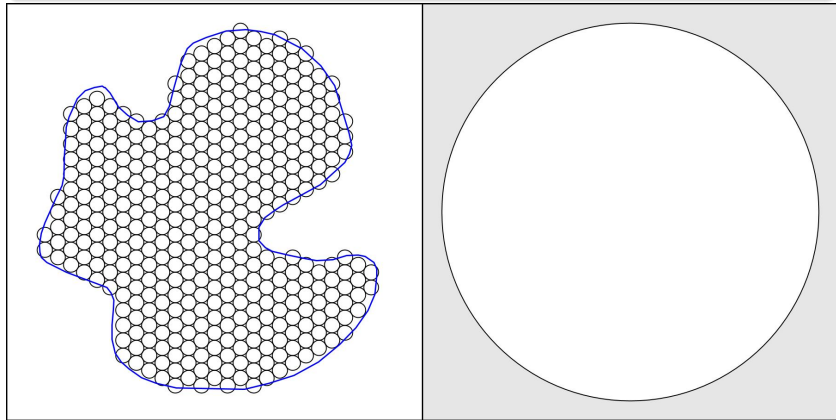
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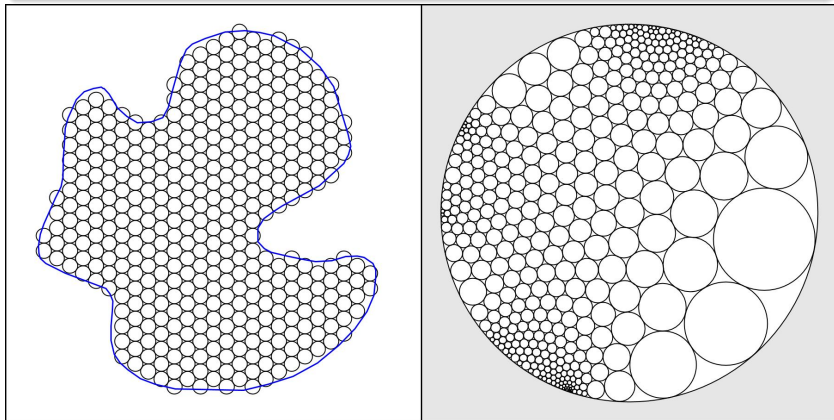
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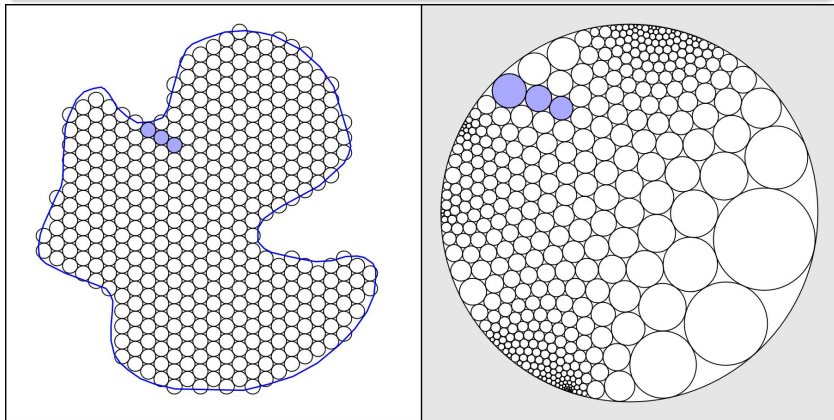
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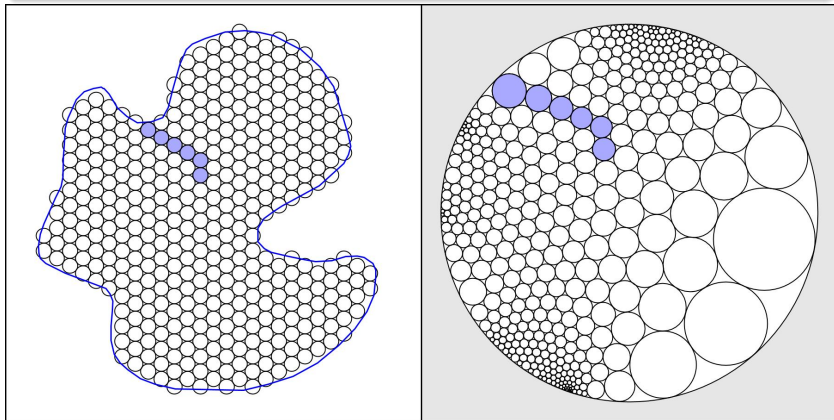
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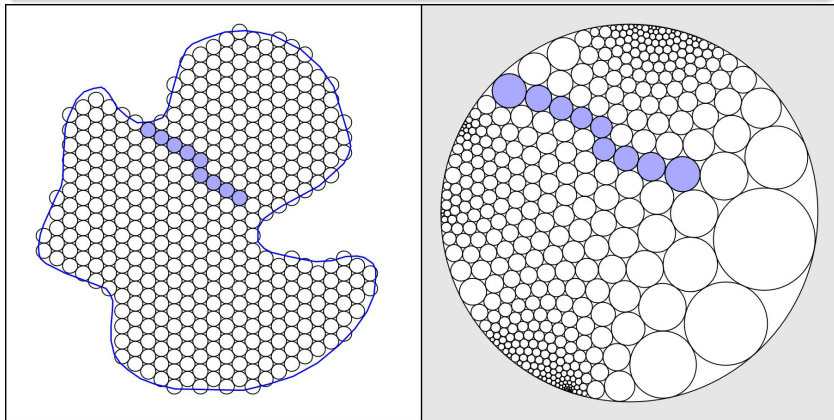
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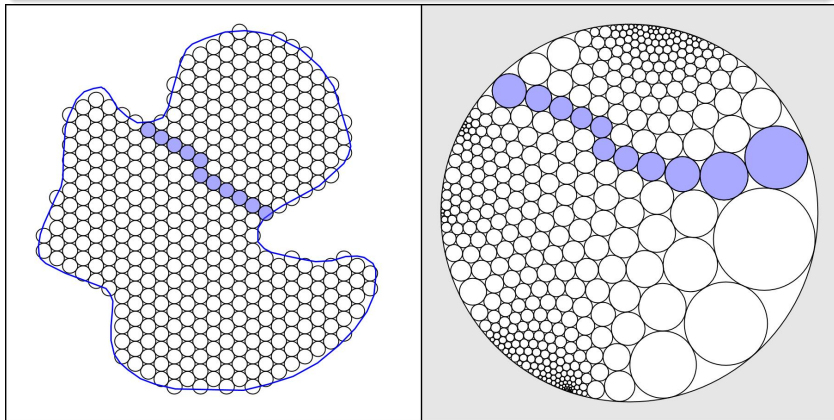
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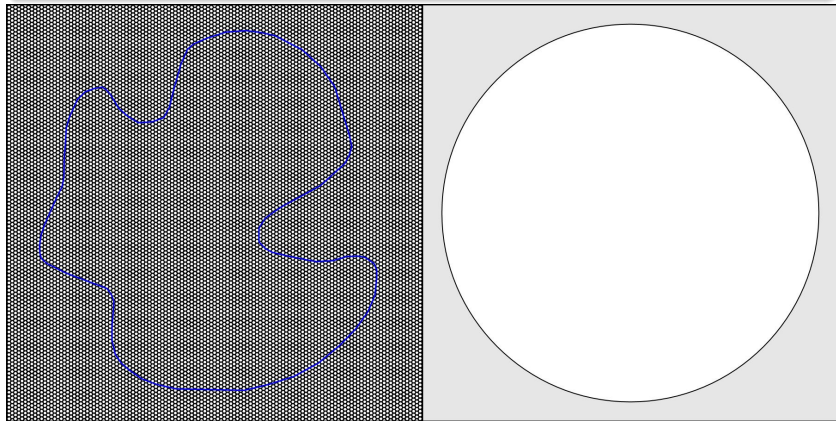
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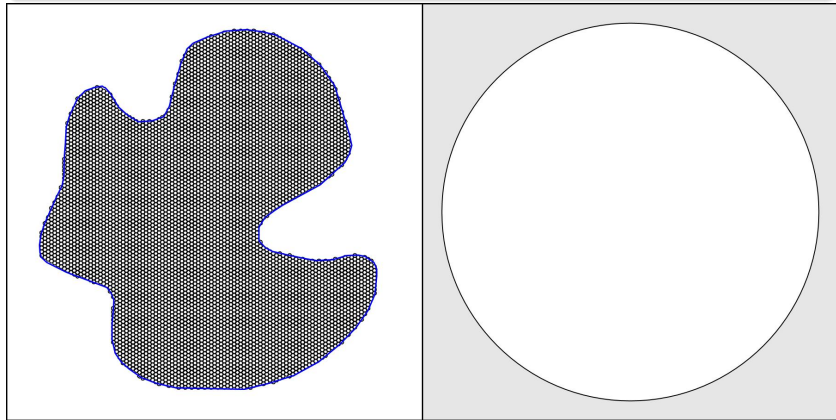
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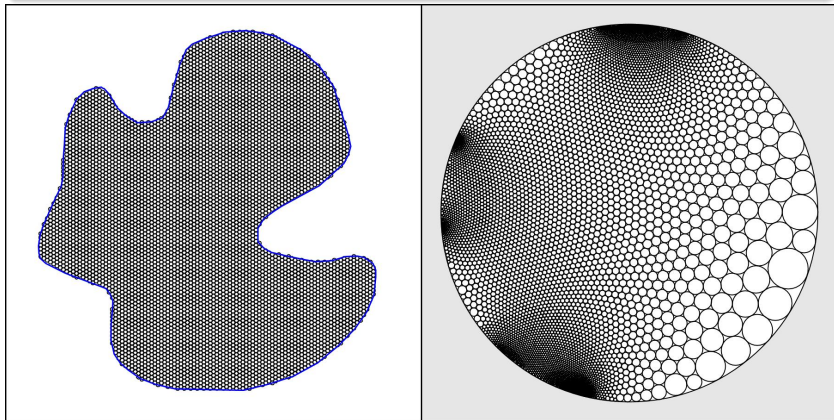
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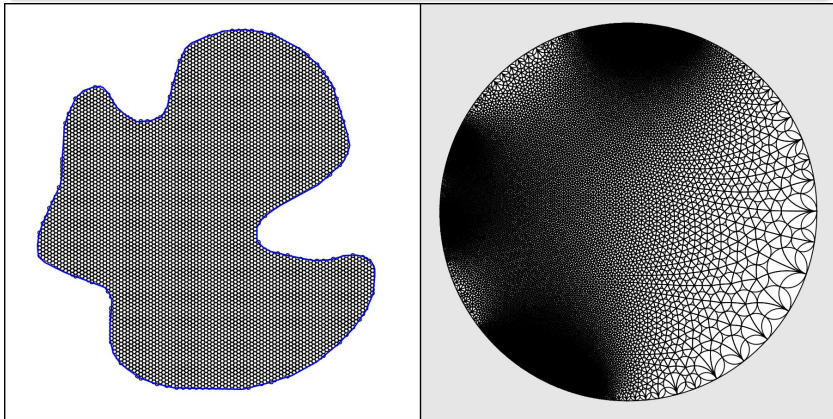
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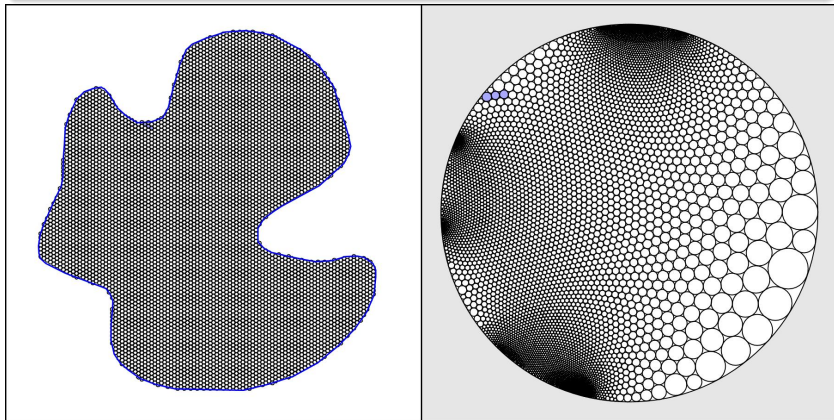
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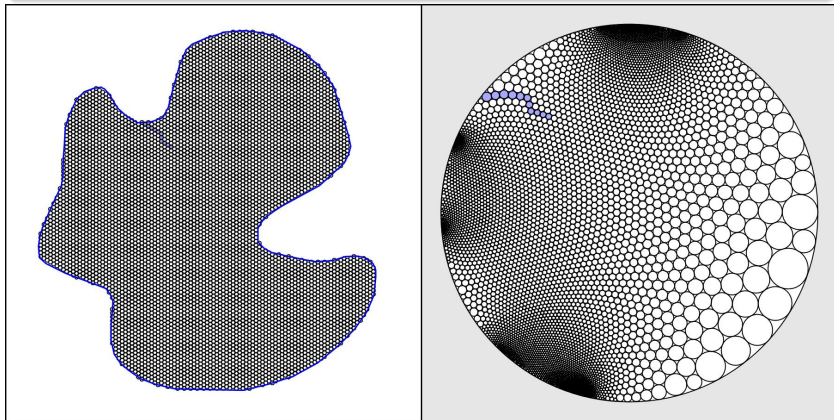
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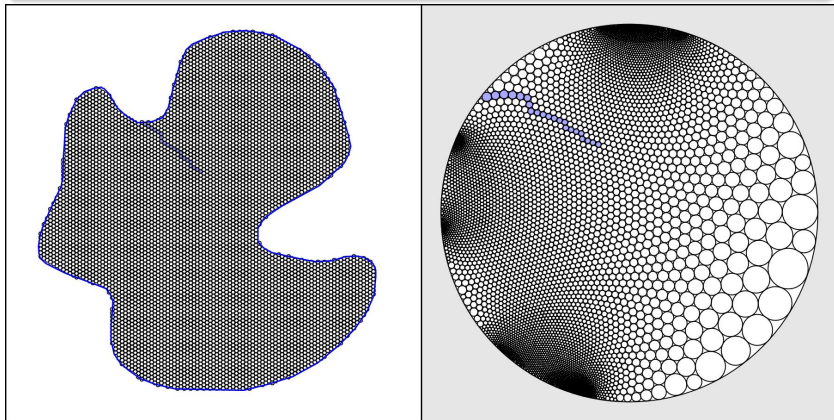
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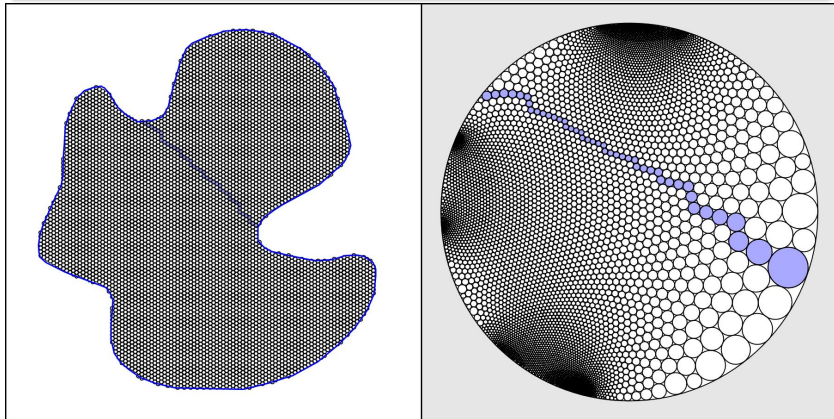
If C is a simply connected finite circle packing in $\mathbb{C}$, then there exists an essentially unique maximal packing of with the same pattern in $\mathbb{D}$.



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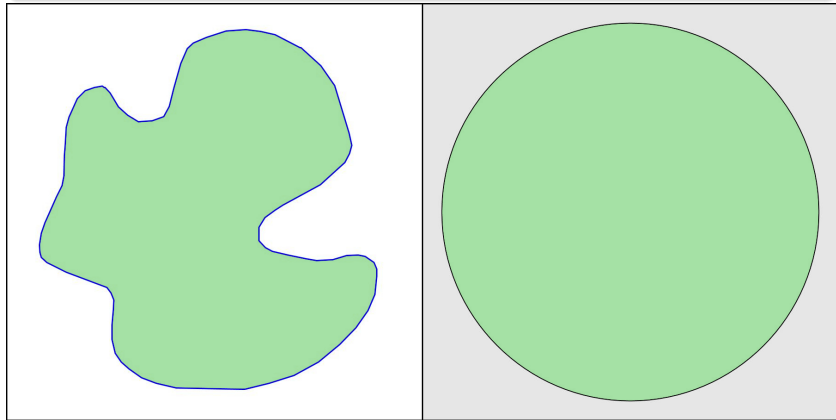


- More circles, and the maps converge to the classical mapping.

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Discrete analytic functions converge under refinement to their classical counterparts.

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[He and Schramm] The KAT theorem implies the classical Riemann Mapping Theorem

Discrete _____ Function

- Discrete Schwarz Lemma
- Discrete Blaschke products
- Discrete polynomials
- Discrete rational functions, etc.

Theorem

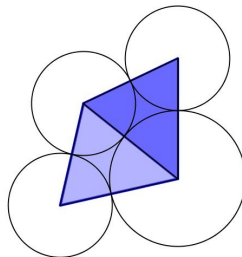
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Branching

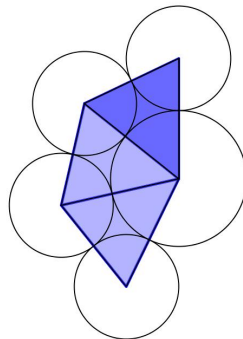
Flowers with angle sums of

- 2π , a (trivially) branched flower.



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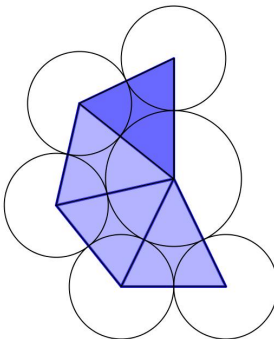


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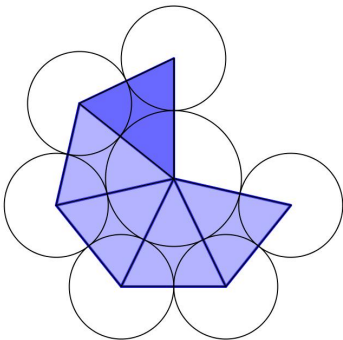
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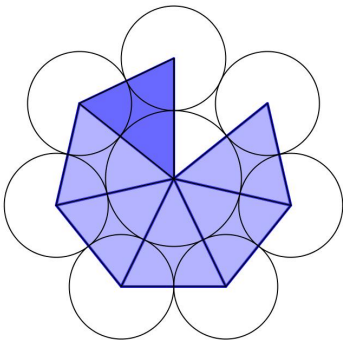
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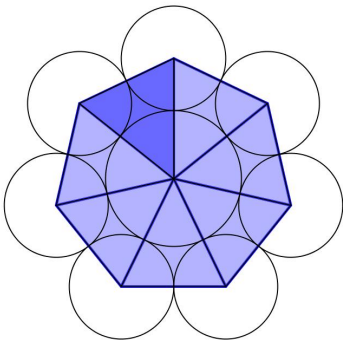
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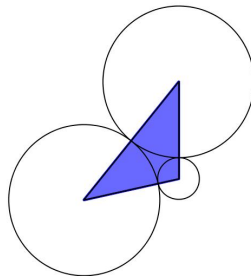
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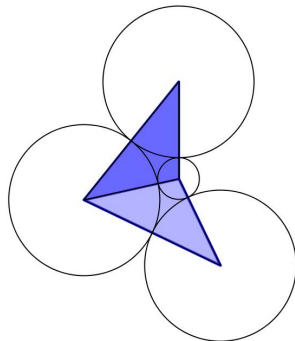
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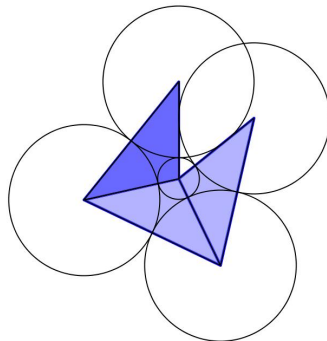
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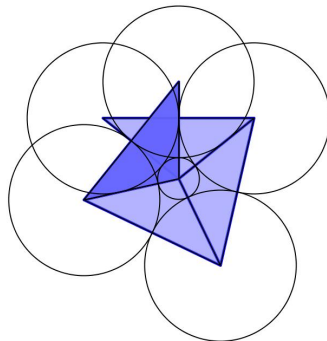
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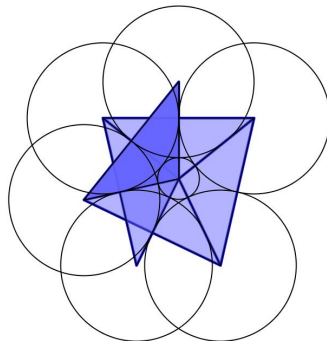
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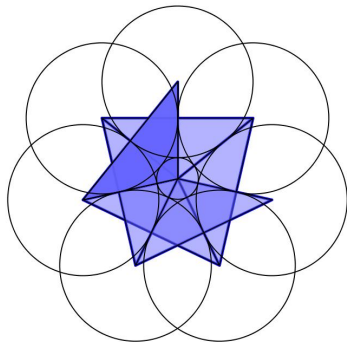


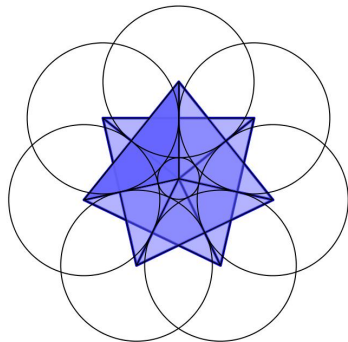
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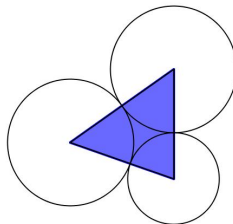
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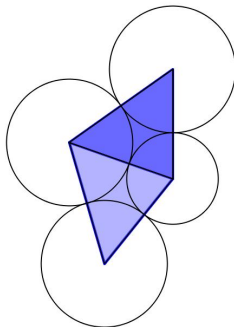


Flowers with angle sums of

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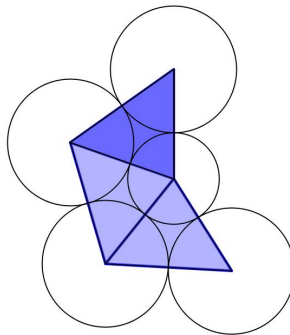
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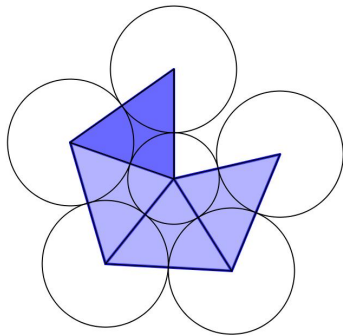
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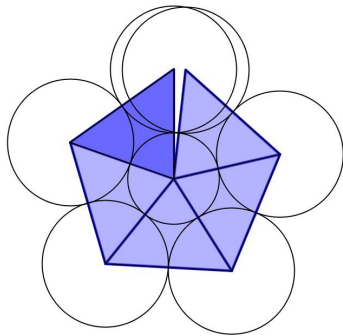
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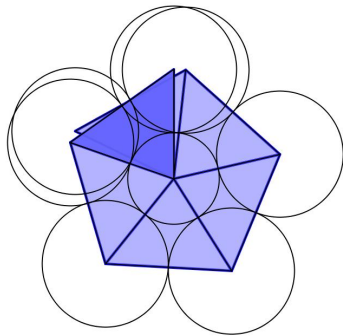
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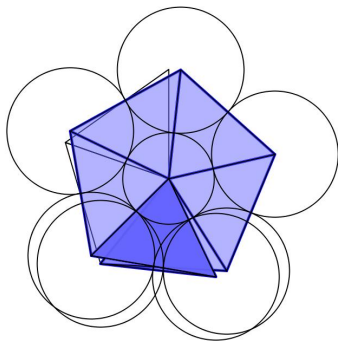
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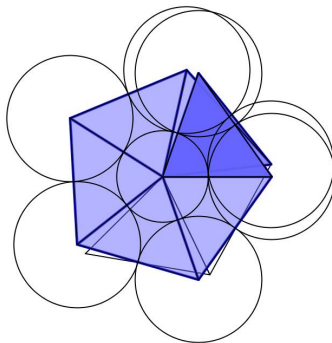
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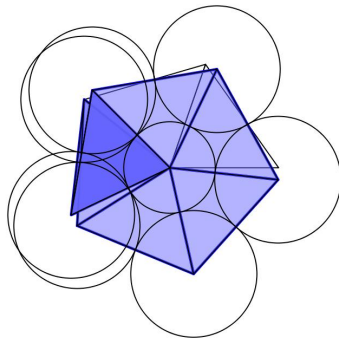
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Interior flowers with angle sums of

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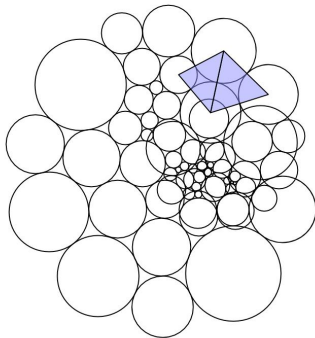
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Branching

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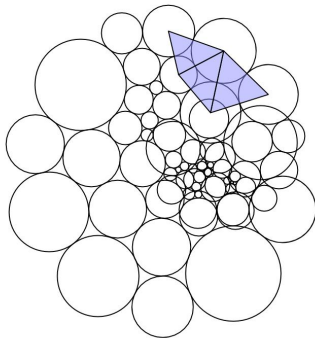
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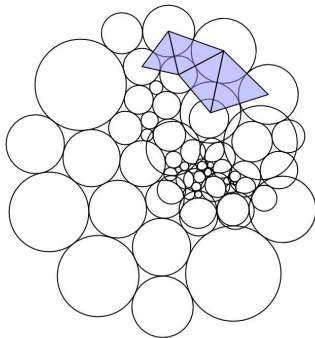
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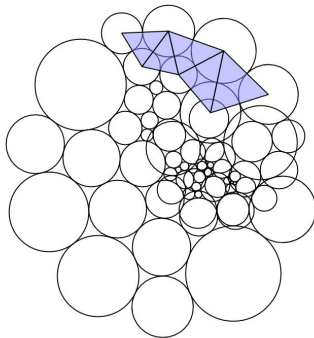
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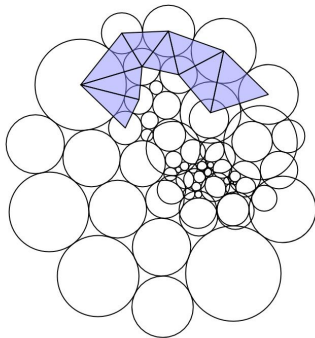
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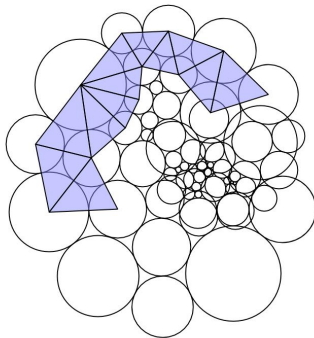
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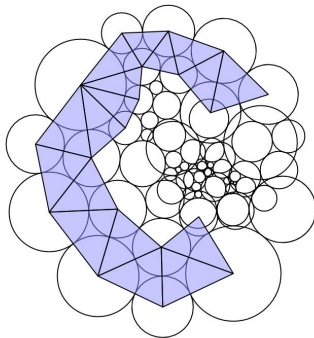
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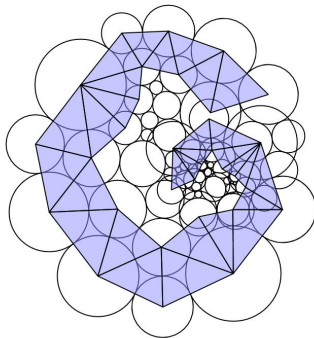
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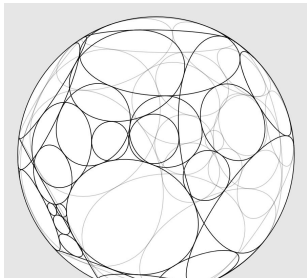
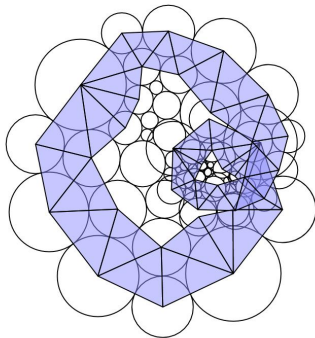
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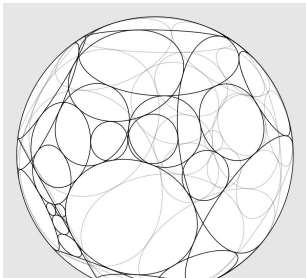
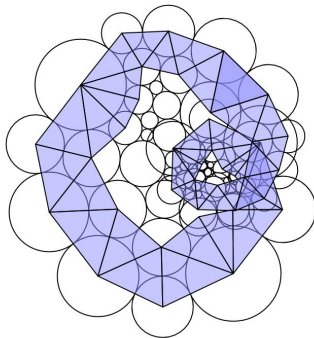
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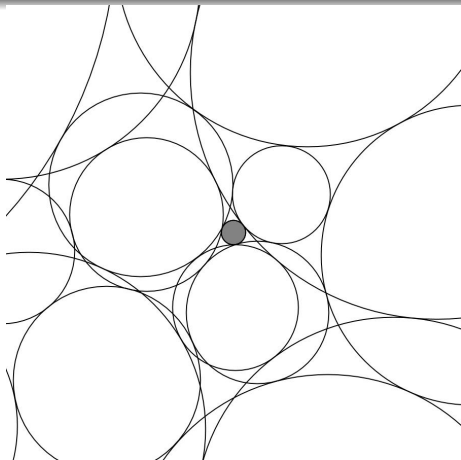
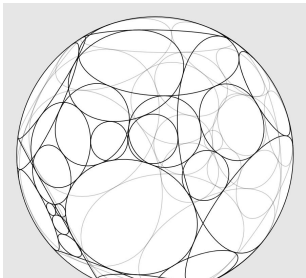
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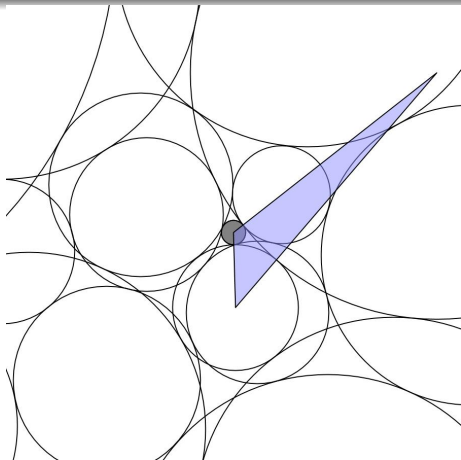
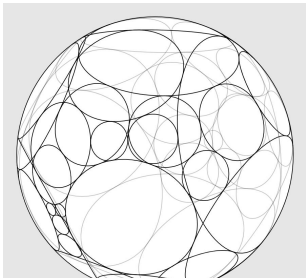
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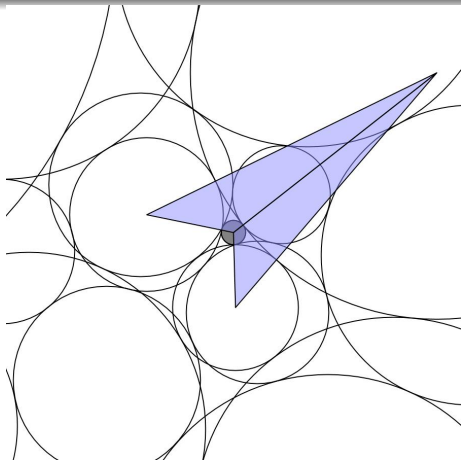
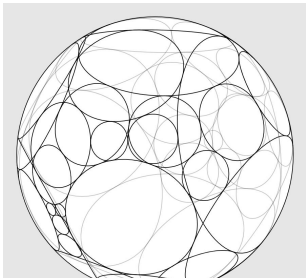
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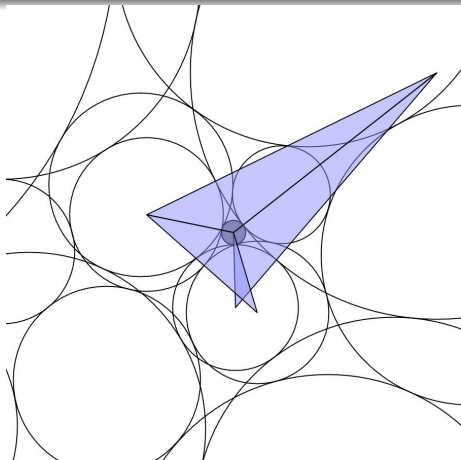
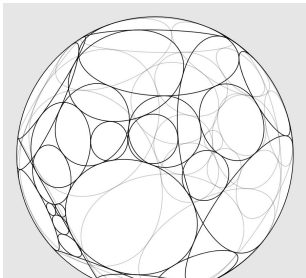
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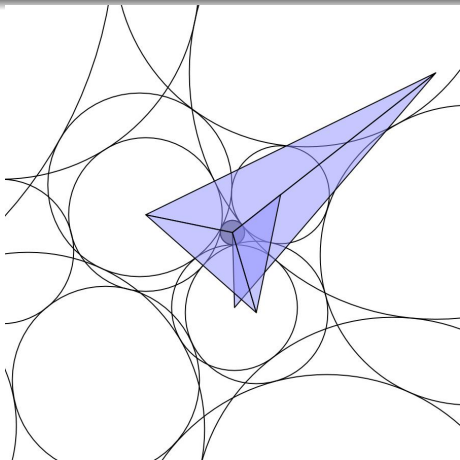
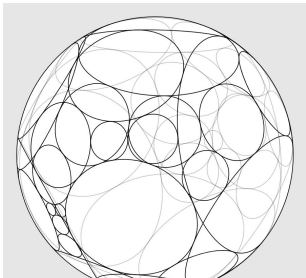
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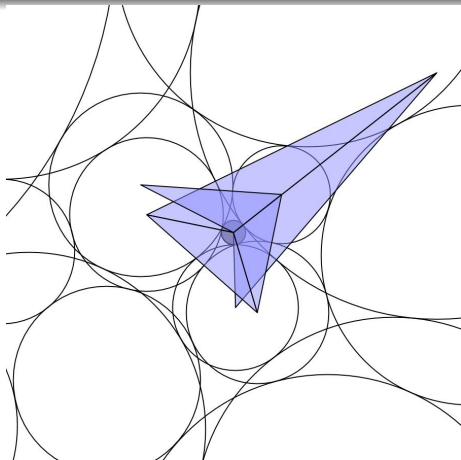
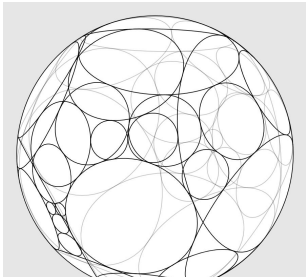
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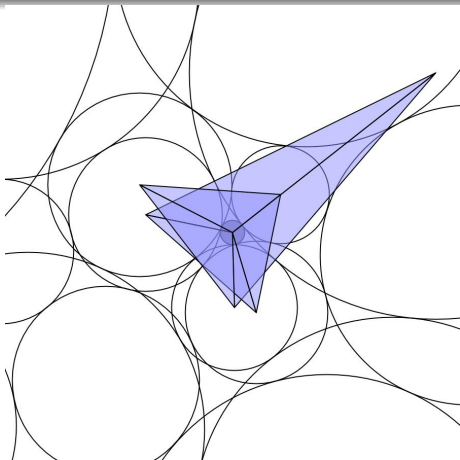
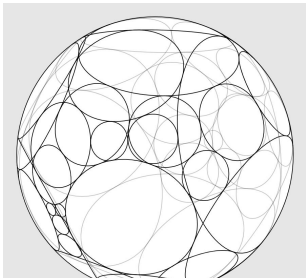
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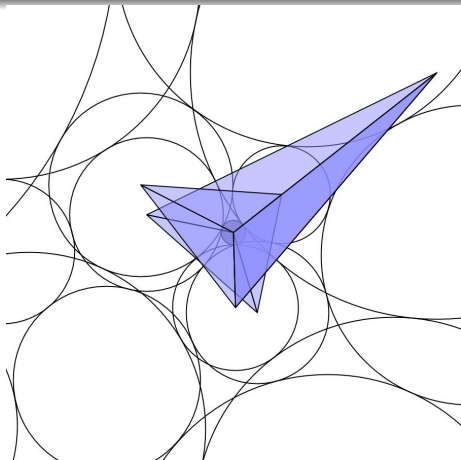
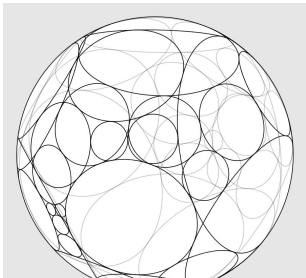
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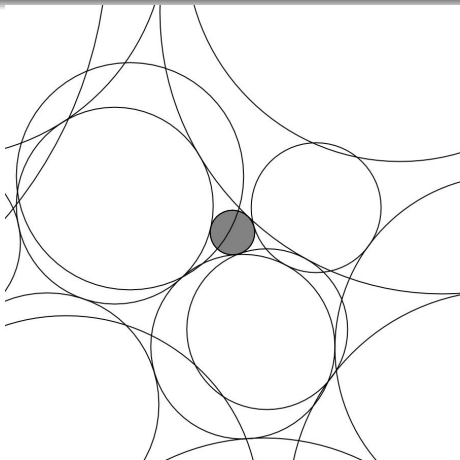
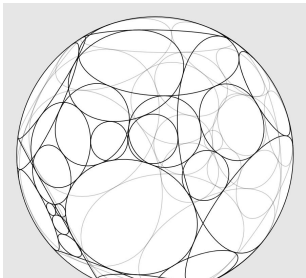
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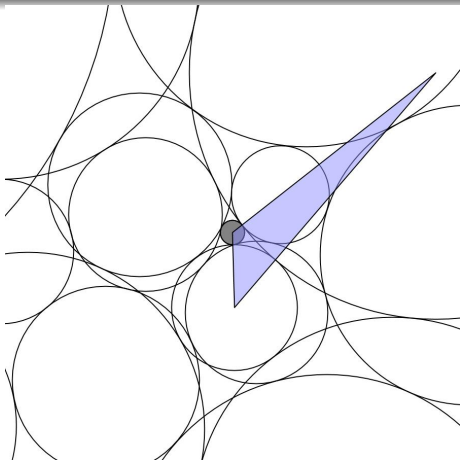
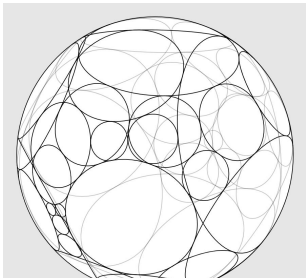
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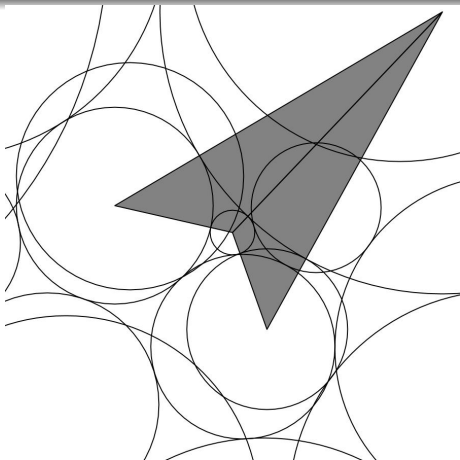
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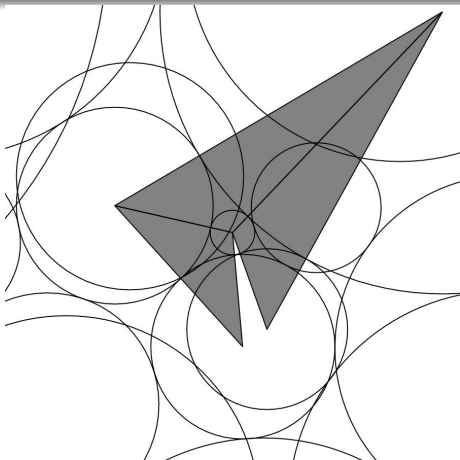
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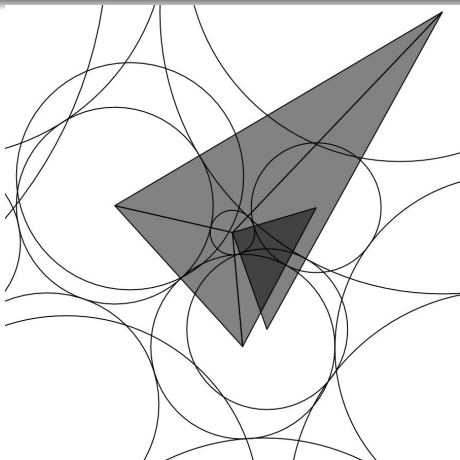
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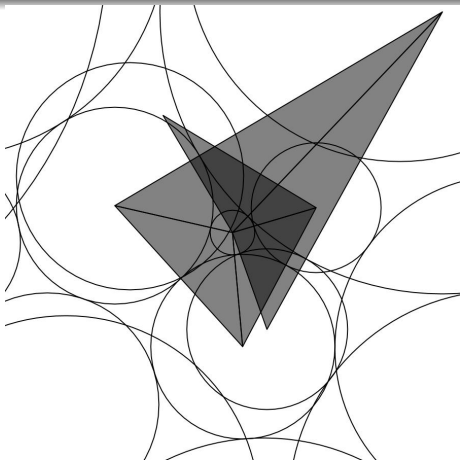
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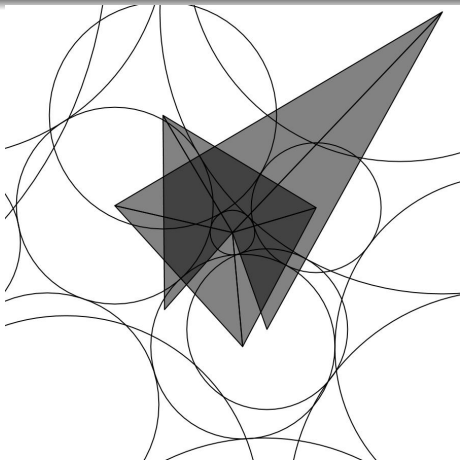
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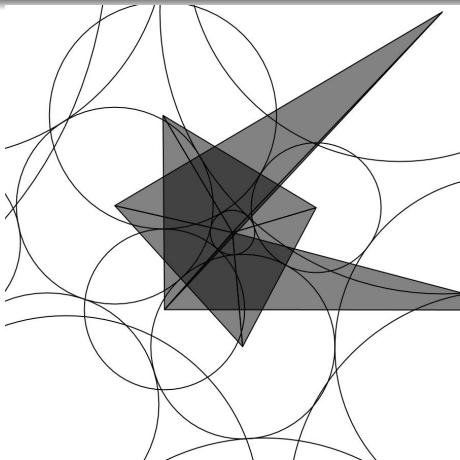
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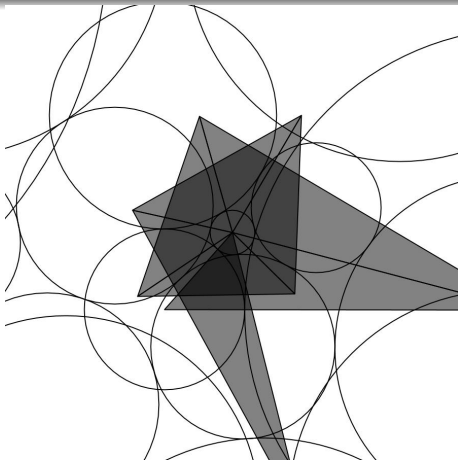
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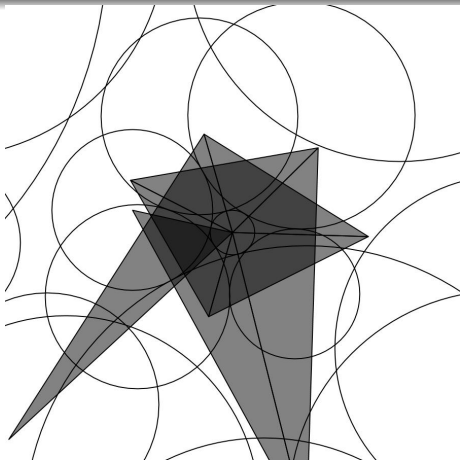
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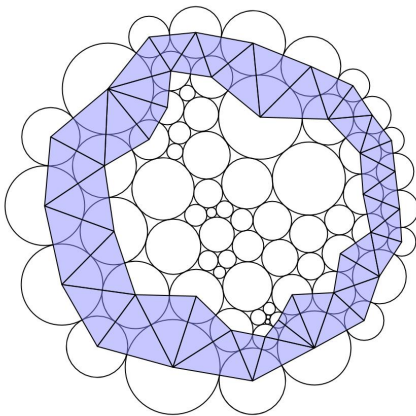
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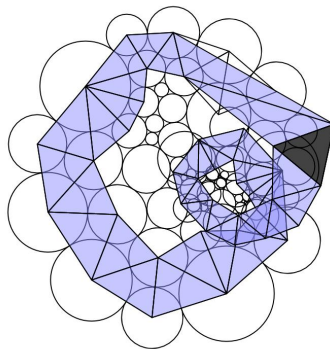
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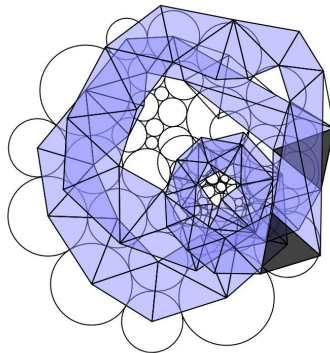
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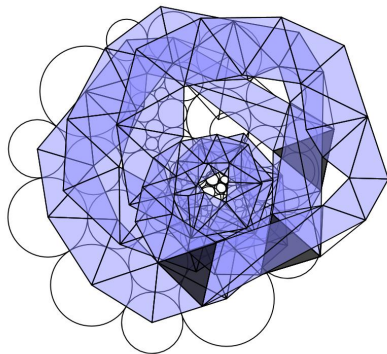
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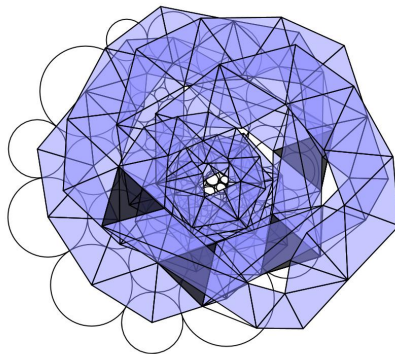
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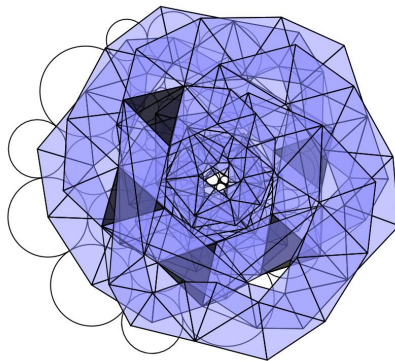
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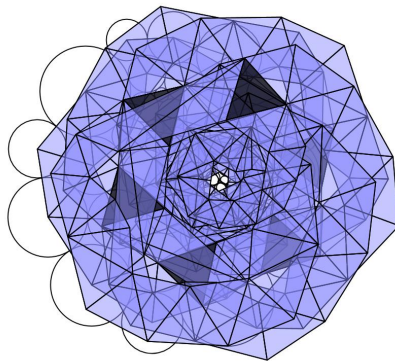
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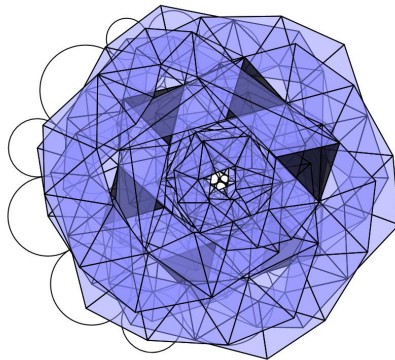
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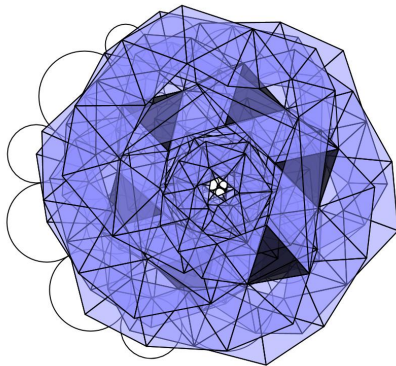
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Thomson Problem

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How to find the min. energy distribution of N electrons on the surface of a sphere?

$$E = \sum_{i < j} \frac{1}{|p_i - p_j|}$$

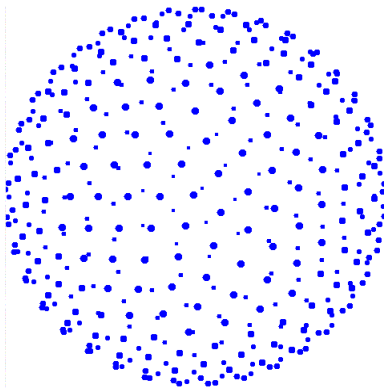
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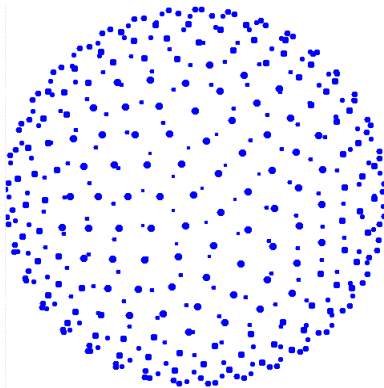


Thomson Problem

How to find the min. energy distribution of N electrons on the surface of a sphere?

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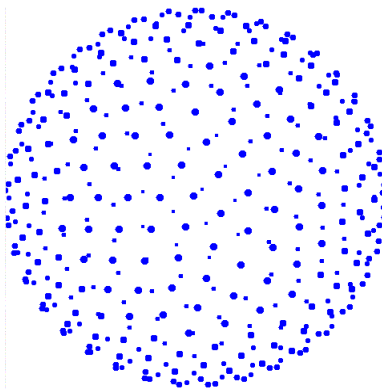
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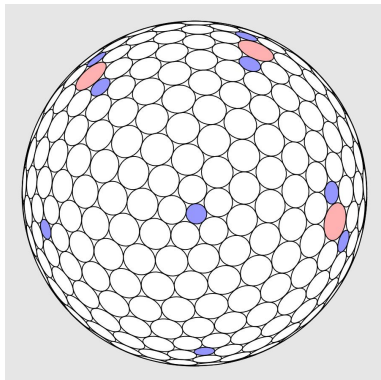
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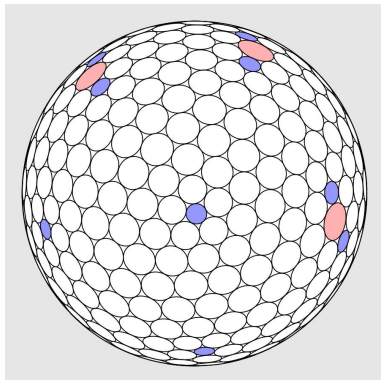
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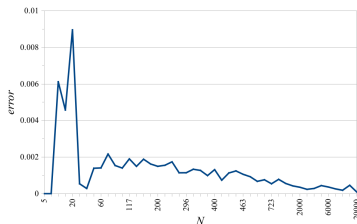


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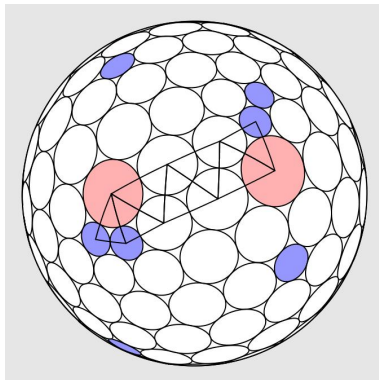
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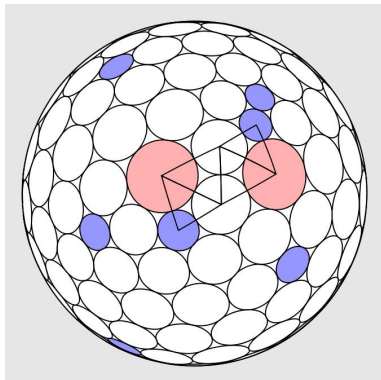
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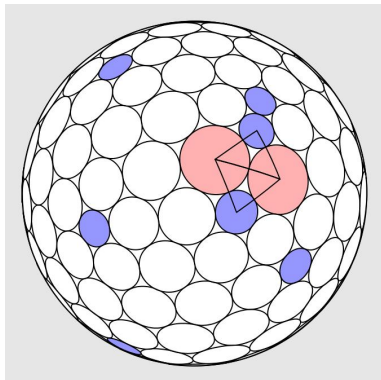
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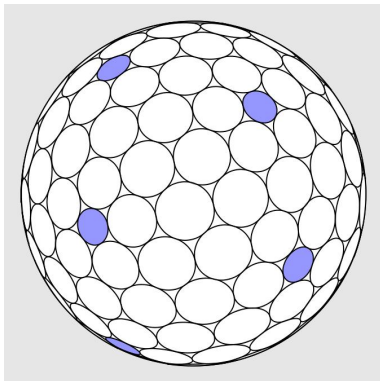
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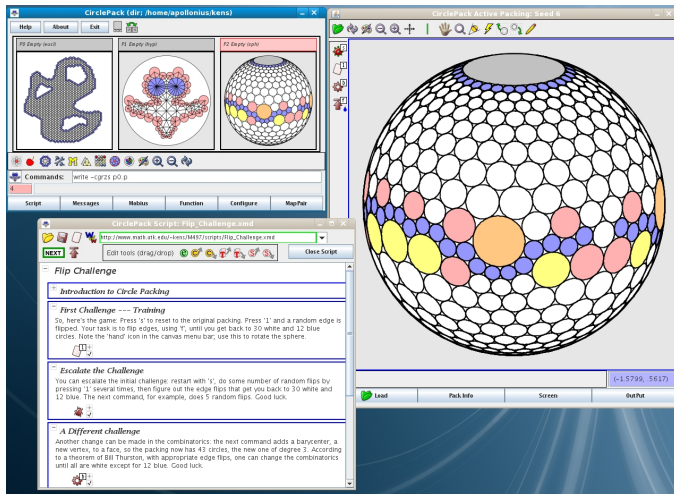
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All images created with CirclePack

<http://www.math.utk.edu/~kens/CirclePack/>



Thanks!